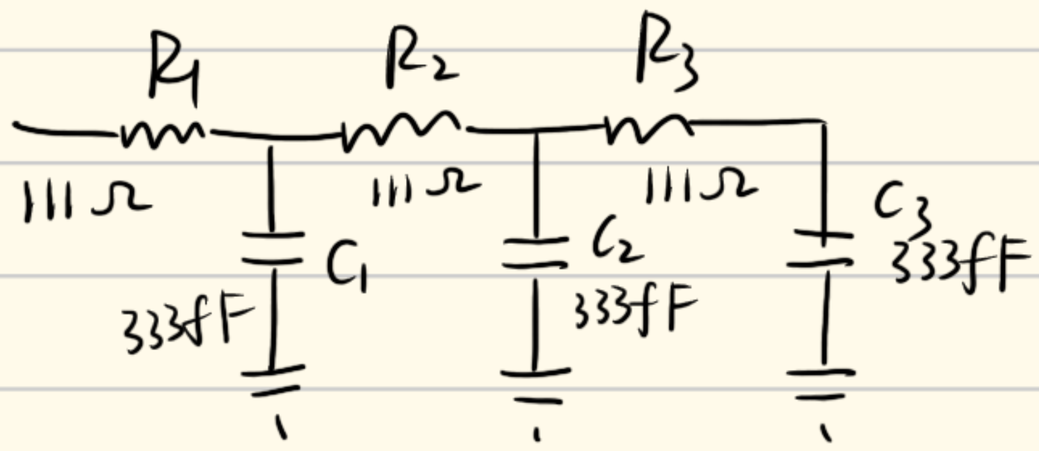
# Exercise 9

## 9-1

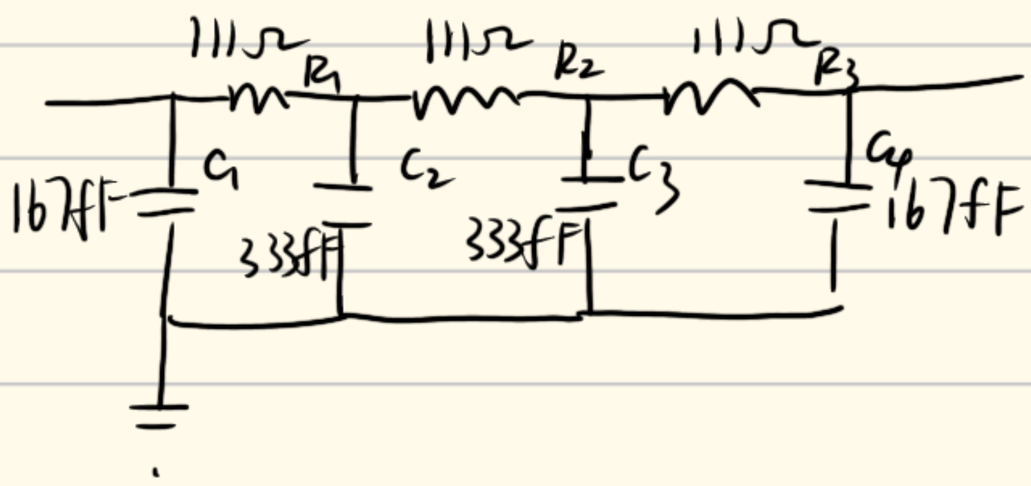
9-1 Consider a 5 mm-long, 4 λ-wide metal2 wire in a 0.6 μm process. The sheet resistance is 0.08 Ω/□ and the capacitance is 0.2 fF/μm. Construct 3-segment L-model and 3-segment π-model for the wire. (Please present models and calculate delay).

① 3-segment L-model




$$\tau = C_1R_1 + C_2(R_1+R_2) + C_3(R_1+R_2+R_3) = 2.22\times10^{-10}\,s$$

② 3-segement π-model




$$\tau = C_2R_1 + C_3(R_1+R_2) + C_p(R_1+R_2+R_3) = 1.67\times10^{-10}\,s$$

## 9-2

9-2 Assume an inverter is designed with a PMOS/NMOS ratio of 3.6 and the NMOS transistor is minimum size. (W = 0.375$\mu$m, L = 0.25$\mu$m, W/L = 1.5). $VM$=1.25V, $VDD$=2.5V. Please describe the voltage transfer characteristic and noise margins of CMOS inverter. (use $V_{IL}$, $V_{IH}$, $NM_H$, $NM_L$ and g). The parameters shown in table 9.1.

Table9.1

| | $V_{T0}$ (V) | γ ($V^{1/2}$) | $V_{DSAT}$ (V) | K (μA/$V^2$) | λ ($V^{-1}$) |
|---|---|---|---|---|---|
| NMOS | 0.7 | 0.45 | 0.83 | 134 | 0.1 |
| PMOS | -0.8 | -0.4 | -1 | 50 | 0.2 |

$K = \frac{1}{2}\mu C_{OX}$

$V_m = 1.25\,V$, $\therefore I_D = k_n\left(\frac{W}{L}\right)_n (V_{GS}-V_{THN})^2(1+\lambda_n V_m) = 68.4\,\mu A$.

$$g = \frac{-1}{I_D} \times \frac{k_n\left(\frac{W}{L}\right)_n V_{Dsatn} + k_p\left(\frac{W}{L}\right)_p V_{Dsatp}}{\lambda_n - \lambda_p} = -24.93,$$

$$V_H = V_m - \frac{V_m}{g} = 1.3, \quad V_{IL} = V_m + \frac{V_{DD}-V_m}{g} = 1.2$$

$$\therefore NM_H = V_{DD} - V_{TH} = 1.2, \quad NM_L = V_{IL} = 1.2$$